LAB EXPERIMENT
ITA0506-COMPUTER VISION

PAVENDHAN A -192321109

EXPNO:35 Morphological Operation Using Black Hat

Aim

To perform Black Hat morphological operation **on an image using OpenCV.**

Algorithm

1. Read BEN 10.png.
2. Convert the image to grayscale.
3. Create a morphological kernel.
4. Apply Black Hat operation.
5. Display and save the result

CODE:

EXP-35.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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```python
import cv2
import numpy as np

# Read image
img = cv2.imread("BEN 10.png")

if img is None:
    print("Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply Black Hat
black_hat = cv2.morphologyEx(gray, cv2.MORPH_BLACKHAT, kernel)

# Display result
cv2.imshow("Original Image", gray)
cv2.imshow("Black Hat", black_hat)

# Save result
cv2.imwrite("EXP35_black_hat.png", black_hat)

print("Black Hat operation completed successfully!")

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Ln: 31 Col: 0

OUTPUT :

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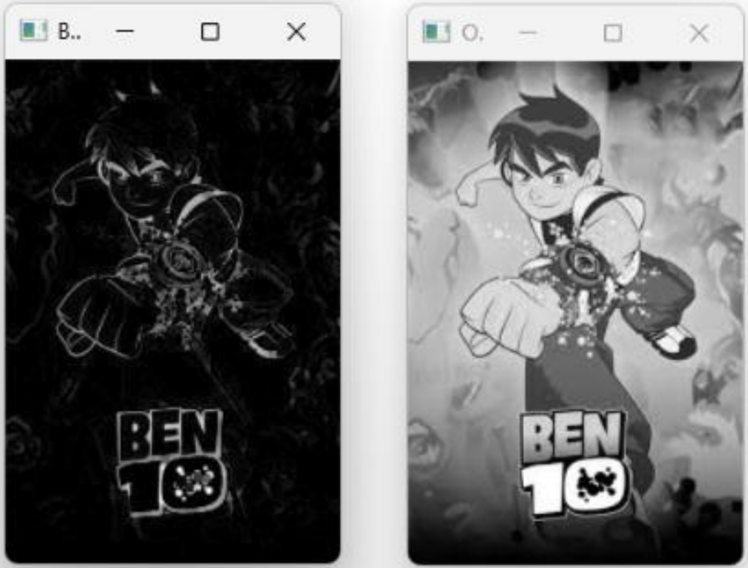
```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab ex
periments/EXP-35.py
Black Hat operation completed successfully!
```



Result

The Black Hat morphological operation is successfully performed to highlight small dark regions in the image.